

Scylla and Charybdis: a small note on the incompatibility between (certain) statistical and the neurobiological models of human language.

Contemporary linguistics, in particular formal syntax seems to me to be facing a double challenge: on the one hand, the revival of statistics based automated discovery procedures such as those exemplified by (v)LLM; for many, these are considered a valid alternative for understanding language structure and language acquisition to the formal methods established in the second part of the XX century by generative grammar(s) (Chomsky et al. 2023 and references cited there); in the other, neurobiologists, both within the subfield of neurophysiology and evolutionary theories, claim that language structure is the emancipated result of language independent modules, such as those stemming from motor and action planning, obviously common to many other animals (Pulvermüller et al. 2010, Lieberman 2010, Corballis 2003). The perilous rout between the Scylla of statistics and the Charybdis of neurobiologists cannot be avoided if linguistics wants to maintain its autonomous scientific force and euristic value.

Given this scenario, at least two consequences appear to me in need to be highlighted: first, the two approaches are not just not converging: they are in fact totally incompatible, since motor and action planning is not acquired by “statistical intelligence” but it is rather driven and restricted by the anatomical and neurophysiological architecture; second, in neither case is the key-notion of “impossible language” relevant (see Moro 2008 for the original proposal and Moro 2016 for a revised one). Assuming “impossible languages” to be conceivable, coherent and perhaps even simpler grammars which are not recognized by the natural networks exclusively dedicated of language, in fact the following twofold conclusion necessarily emerges: for (v)LLM, impossible languages cannot be defined because there are not such things as natural circuits for machines; the case of action and motor planning, instead, impossible languages cannot be defined because language structure is claimed to be directly linked to the degree of control of movement allowed by the anatomy of the human body and as such if they are conceivable they must also be possible (Moro et al. 2023).

There has been an intense and – at least to me - sometimes ideological and myopic discussion about the role of statistical methods to understand human languages (plural matters here), witness the amount of papers and debates on (v)LLM. But on contrary there has been a relatively quite scarce attention to the fact that the other side of the vision on language has received far less attention, especially when it comes to its inconsistency with the former one. The following two short papers, reproduced here, the former of which has never been available on this platform, clarify why the “action-motor emancipated theories” fail to be plausible models for language and offer theoretical and empirical reason to why

these two approaches are incompatible. It's in between them that the force and the autonomy of linguistics should receive new support to be (re) established as a fruitful domain of scientific inquiry.

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On the similarity between syntax and actions

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One of the major discoveries in the history of 20th century linguistics is that the linear sequence of words constituting a sentence is organized in a hierarchical and recursive fashion. Is this hierarchical structure similar to action and motor planning, as recent proposals suggest? Some crucial differences are highlighted on both theoretical and empirical grounds that make this parallel unsuitable, with far-reaching consequences for evolutionary perspectives.

Apparent similarity between language and actions

One of the major discoveries in the history of 20th century linguistics is that the linear sequences of words (the lexicon) is organized in a hierarchical fashion via a simple binary recursive computational operation that merges lexical items (technically referred to as Merge). Is this architecture specific to human language structure or are there analogous patterns in other cognitive domains? The answer to this question has very far-reaching consequences, including the way in which we can search for evolutionary clues concerning the human mind.

The search for neuropsychological networks underlying this architecture is one of the major empirical and theoretical challenges of contemporary neuroscience. Recent work has paved the way towards this goal [1,2], but some major issues still need to be clarified [3–5]. Moreover, not all hierarchical structures are necessarily recursive in human language, such as the syllable structure, which makes syntax special with respect to other modules of grammar. Notably, it is this hierarchical architecture that matters for any syntactic relation, and not the linear order, which is just an epiphenomenon for computational, formal, and neuroanatomical reasons [6–8], including nested dependencies. A typical example at the word level is the ‘either/or if/then’ dependency, but dependencies can typically occur even below the word level, such as in the ubiquitous case of agreement morphemes. Concentrating on words, when a word such as ‘either’ enters the computation, a dependency is established with ‘or’ at two points of the hierarchy, and hence of the linear sequence. Analogous dependencies can be nested *ad infinitum*, provided that memory load and parsing restrictions do not intervene to block it, as in the

following example, where A and B denote the correlated elements: [either_A someone told you that [either_B you go to Rome or_B you will be arrested] or_A you are a fool to go there].

One study addressed the hypothesis that ‘basic body acts are joined in action chains to form a meaningful goal-directed action sequence’ and that ‘the hierarchical structure of embedded or “nested” sentences is paralleled (by them)’ [9]. In other words, *mutatis mutandis*, the sequence of actions represented as [open the door [switch on the light] close the door] would parallel the former type of syntactic dependence established in ‘either/or’ sentences, suggesting that actions are organized according to the same architectural model as the syntax of human language. Is this analogy between language and actions true or, at least, useful? The implications of this hypothesis are very far-reaching because they obviously have a bearing on evolutionary considerations.

Actions take place in the physical world and are subject to its physical restrictions (such as particle interactions and anatomical structures); by contrast, syntax is essentially a mind/brain phenomenon for which the physical conditions are relevant only insofar as the brain is itself a physical object. Certainly, actions can be organized hierarchically, as can be the branches and leaves on a tree. However, syntax does not only consist of a lexicon and Merge; it also crucially includes a filter limiting the number of successful combinations yielded by Merge, technically called (principles of) Locality. Does the ‘syntax of actions’ meet these three requirements? The analogy between syntax and actions must accordingly be checked against the notion of ‘word’ and Locality to test if it is empirically tenable. We first consider the notion of ‘word’.

Does the notion of ‘word’ apply to actions?

On the basis of studies in macaque monkeys, it has been proposed that a ‘motor vocabulary’ indeed exists [10]: ‘This motor vocabulary is constituted of “words” [...] These words select specific “motor prototypes” such as, for example, the configuration of fingers necessary for the precision grip.’ There can be hardly any doubt that the idea of a set of ‘preformed motor prototypes’ is very appealing. Is it also true when compared to the structure of human language?

One of the universal properties of human languages is that the lexicon can be parted into two classes: an unbounded set of words carrying the meaning of objects, actions, qualities, and feelings, and a small and fixed repertoire of ‘functional’ words containing ‘logical’ and set-theoretical instructions (such as ‘the’, ‘if’ or ‘not’). If there are no pathologies, such as in agrammatism, the burden of information is practically equally distributed

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between these two classes in every sentence. If the ‘motor vocabulary’ is constituted of ‘motor prototypes’, then there is no hope of having anything equivalent to functional words in the vocabulary of actions. Of course, we do exploit functional words to describe a motor action verbally, as in ‘open your hand if you want to grasp the knife’, but this would just take us back to the structure of language. At the level of the lexicon, then, the analogy between language and actions does not seem to be fully suitable. However, the most problematic aspect for the analogy between the syntax of words and actions comes from the notion of syntactic dependency.

Syntax contains a filter based on hierarchy: can there be one for actions?

A recalcitrant puzzle of human language stems from the contrast between the fact that there is no upper limit to the length of a sentence due to recursive procedures and the fact that successful combination of lexical items appears to be strictly limited across languages by Locality conditions [11,12]. Typical examples of these limits affect so-called displacement, a ubiquitous phenomenon that has been recognized since the first decade of the last century. A prototypical case is question formation in English, whereby the dependency between, say, a verb and its object may be established even if the two are not adjacent. For example, the dependency between ‘book’ and ‘read’ in the complex sentence [Peter thinks [that Mary read a book]] is the same as in [which book does Peter know [that Mary read]?], even though the closer potential verb on which ‘book’ could depend is ‘know’, as in the sentence ‘They know a book.’ How does Locality restrict displacement? Consider for example the following two sentences: [[Peter saw Mary reading a book] [before writing the essay]] and [[Peter saw Mary writing the essay] [before reading a book]]. Surprisingly, the dependency between ‘reading’ and ‘book’ can be established only in the first sentence after displacement takes place: [[which book did Peter see Mary reading] [before writing the essay]?] versus [[Which book did Peter see Mary writing the essay] [before reading]?]. Memory-buffer-based solutions to these types of contrasts fail, as demonstrated by the possibility of establishing dependencies at a longer distance in the sentence [which book does John think [my father’s blonde sister believed [Peter saw [Mary reading]]?]].

Locality has been at the very center of theoretical syntax since it was first discovered and is recognized as a core and definitory component of syntax that massively reduces the amount of computation: in no human language can Locality be absent. What matters here is that this filter is essentially geometric in the sense that it is based on the hierarchical relations resulting from Merge. Explanation of the contrast under discussion here, for example, is based on the fact that in the ungrammatical case, the sentence containing the verb ‘reading’ is merged after the first one is already fully assembled, as indicated by the brackets. In fact, both Locality and displacement can be (partly) derived as a consequence of Merge if a Dynamic Antisymmetry approach to movement is adopted [7,8]. Returning to the original claim that a sequence of actions verbally

expressed, such as [open the door [switch on the light] close the door], is governed by the same principle as syntax, we realize that unless some Locality principles are found that are based on the alleged hierarchical and recursive structure connecting actions, no parallelism can be inferred. This is in fact a completely open and challenging question. No commitment appears to be empirically and theoretically legitimate unless this preliminary fact has been clarified.

Nothing more than an (appealing) metaphor

If the Merge operation could be generalized to intentions and motor planning actions yielding the same rich set of complex hierarchical recursive structures and if Locality conditions based on them could be observed in this realm as well, then the analogy between syntax and the domain of actions may become more than a metaphor and we could start to explore the possibility of finding a precursor to the basic operation of composition yielding all properties characterizing human syntax.

However, the failure of this analogy can be regarded as welcome. If motor actions were the phylogenetic and ontogenetic precursor of human language [13], the singularity of this code across species, most crucially with respect to monkeys, would be a much harder mystery to solve [14,15], especially because the specific format of human syntax has been definitely anchored to the neurobiological structure of the brain [1,2,14].

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Response to Pulvermüller: the syntax of actions and other metaphors

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Let us first consider the notion of embedding in syntax with a simple example as in ‘the men [who he saw] were tall’. We have a phrase in square brackets [who he saw] with a specific structure and a specific distribution (i.e., it can appear in certain contexts but not in others); this phrase is embedded in another structure [the men were tall] where a dependency is established between ‘men’ and ‘were’. In fact, the verb ‘be’ cannot enter in a relationship with any other element of the sequence but ‘men’; if it did with the closer element ‘he’, as in ‘he was tall’, the sentence would be ungrammatical – ‘the men [who he saw] was tall’. Of course, not all ordered sequences involve embedding and dependency. Take for example the first letters of the alphabet, ‘abcdefg’. Although one can isolate a subpart of it – say the sequence [cde] – this subsequence is not endowed with a specific structure nor restrictions based on the context of distribution; moreover, there is clearly no dependence between the element and where the sequence has been embedded [ab...fg], unlike the case of the sentences above. The sequence could well be ‘abcdehg’.

Let us now turn to actions, as in the case observed by Pulvermüller: [open the door [open the bottle close the bottle] close the door]. What we see here is analogous to the sequence of letters in the alphabet rather than the sequence of words in the sentence: the action [open the bottle close the bottle] does not have any special distribution and there is no dependency between [open the door] and [close the door], which is the structure where [open the bottle close the bottle] is ‘embedded’. In fact, there is no embedding at all; rather, there is just a sequence (although, of course, the actions may be connected in a causal way). Suppose that the door is built in the middle of a wall that stands alone in a garden. I can open the door and go through it without coming back and closing the door; indeed, [close the door] in the sequence under discussion can be substituted by a virtually infinite number of different types of action [[sing *Aida* backwards], [jump], [open another door]] yielding, for example, [open the door [open the bottle close the bottle] jump], unlike the case of verb agreement we observed. Moreover, in human languages dependencies can be nested as in ‘if [the men [who he saw] were tall] then it was raining’. The agreement dependency we observed before between ‘men’ and ‘tall’ is

nested in another independent dependency between ‘if’ and ‘then’; that this is a dependency is trivially clear because ‘then’ cannot be substituted by, say, ‘or’, which would rather require a dependency with ‘either’. Notice that embedding goes on without bound in human language. Consider this simple example: let *S* be a sentence containing *n* nested dependencies and *T* a sentence; ‘if *S* then *T*’ is a grammatical sentence but ‘if *S* or *T*’ is not. All humans know this without explicit instructions or training and this has been proven to be mathematically beyond the bounds of finite automata, as observed 60 years ago [1]. Regarding questions related to memory limitations, note that it is well known that, without external aids, people can understand such examples with up to about seven nested dependencies [2]. This is the behavioral correlate of short-term memory limitations. Finally, note that if we refer to ‘usage’ as a theoretical notion to uncover the capacity of the human mind to deal with a code and apply it to arithmetic, we are led to the unacceptable consequence that knowledge of arithmetic is finite, because without external aids people are simply not able to perform unbounded operations.

In summary, disregarding the absence of an analog of functional words and locality conditions for actions (syntactic movement not being a special case at all), the idea of ‘a syntax of actions’ remains a metaphor if compared with the syntax of any human language. Perhaps, as noted in the paper, if we shift from actions to action planning some analogies with syntax could be explored, although it remains to be proved that this activity is not essentially parasitic on language [3]. Given that animals seem to be able to plan actions, I still consider this separation between actions and syntax a welcome fact for any research on the human brain and evolution [4–7]. Once more, no commitment appears to be empirically and theoretically legitimate, unless these preliminary issues have been clarified.

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